

Michele De Vita

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🔄 Michedev

🏠 micheDEV.github.io

AI Safety expert and PhD candidate working on the reliability and robustness of deep learning systems, from distribution-shift detection to uncertainty estimation for safety-critical applications.

EXPERIENCE

FAU Erlangen-Nuremberg
PhD Candidate 09/2023-now

- Working with Mercedes for Trajectory prediction and with Nokia for predictive maintenance
- Designed and taught the lab course of Machine Learning And System, where the students had to deploy pre-trained Deep Learning models to edge hardware i.e. Jetson Nano and Raspberry Pi

Freelance Online
Teacher 2017-2023

- Machine Learning courses for three big Italian companies managers: Poste Italiane, Alfamotion and BeeVoip
- Taught Data Science and Deep Learning to over 50 workers and students
- Mentoring website page (automatically English translated)

Prediba Copenhagen
Co-founder and CTO 2019-2020

- Prediba is a start-up that offers predictive maintenance to manufacturing companies using AI
- Responsible to communicate to investors and partners our solution
- Managed two freelance software developers

Awards

- #366 at Google Hash Code coding competition - [link](#)

EDUCATION

University of Florence Florence
Master's Degree in Data Science 10/2017-10/2022

- Final grade: 110 cum laude/110
- Courses: Deep Learning, Machine Learning, Statistical modeling, Bayesian Inference, Parallel computing

DTU Copenhagen
Erasmus Master's Thesis 09/2021-03/2022

- Master Thesis on Unsupervised Object Centric representations supervised by [Ole Winther](#).
- Published a paper based on Master's Thesis work
- [Thesis link](#)

University of Florence Florence
Bachelor's Degree in Computer Science 09/2013-07/2017

- Internship for bachelor's thesis in Indoor positioning using smartphone magnetometer

PUBLICATIONS

Forecasting the Past: Gradient-Based Distribution Shift Detection in Trajectory Prediction 2026
[CVPR \(CVPRW\) 2026](#)

Diffusion Model Guided Sampling with Pixel-Wise Aleatoric Uncertainty Estimation 2025
[WACV 2025](#)

Parameterization Learning with Convolutional Neural Networks for Gridded Data Fitting 2024
[FICC 393-412, 2024 Outstanding Research Award](#)

Generalization and Robustness Implications in Object-Centric Learning 2022
[ICML \(ISSN: 5221-5285\) Vol 162, 2022](#)

SKILLS

Hard skills

Programming - Python, C++, Java - **Deep Learning** - Generative models, Transformers, PyTorch, VAE, DDPM - **Web Client Development** - Javascript, Angular, HTML, CSS - **System and tools** - Docker, Linux, Git, CI, SSH

General

Language Skills - Italian as mother tongue, fluent English, basic Spanish , **Interests** - Painting, reading, open sourcing